

Paper IV - Recent Advances (MCh Urology, Summer 2015)

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IV-1. Role of robotics in urology

Type: LAQ (25 marks)

Introduction / Definition

- Current surgical robots are **master-slave telemanipulators**: the surgeon controls every movement, so the robot is not autonomous.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, which is why RARP (robot-assisted radical prostatectomy) spread where pure laparoscopic prostatectomy had stalled. RARP is now the dominant approach in the USA.

History (milestones)

Year	Milestone
1985	PUMA 560 : first surgical robot (brain biopsy); PROBOT (late 1980s) was the first urological robot
1997	First human trial of the robot later developed into da Vinci ("Mona")
2000	da Vinci FDA (Food and Drug Administration) approval; first robotic radical prostatectomy (Binder and Kramer)
2002	First paediatric robotic procedure
2019	Key da Vinci patents expired, so new platforms appeared

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) view magnified about 10 times, master controllers, motion scaling and tremor filtration.
2. **Patient-side cart**: arms carrying the camera and **EndoWrist** instruments with **7 DOF (degrees of freedom)** and 90° articulation; most 8-mm instruments have a 10-use limit.
3. **Vision cart**: light source, camera electronics, energy generator and monitor.

Generation	Added feature
Standard (2000)	Three arms
S (2006)	High definition, fourth arm
Si (2009)	Dual console (training)
Xi (2014)	8-mm camera in any port, side docking, multi-quadrant surgery

Generation	Added feature
SP (2018)	Single port: all instruments through one 2.5-cm cannula
da Vinci 5 (2024)	Force feedback

- **Other platforms:** Hugo RAS, Versius, and the low-cost Indian **SSI Mantra**.

Advantages and disadvantages

Advantages	Disadvantages
3D magnified vision	High capital cost plus limited-use instruments
Wristed instruments, easy intracorporeal suturing	Limited haptic feedback
Tremor filtration, motion scaling, ergonomics	Docking time, bulk, surgeon remote from patient
Less blood loss and shorter stay than open surgery	Learning curve; limited public-sector access

Applications in urology (with evidence)

1. Radical prostatectomy (index procedure)

- **Yaxley et al., Lancet 2016** (RCT (randomised controlled trial), open vs robotic): urinary and sexual function were **equal at 12 weeks**; robotic surgery had less blood loss and a shorter stay. **Coughlin 2018**: still equal at **24 months**.
- Meta-analyses show no difference in positive margins. The surgeon's skill and volume, not the platform, is the main variable.
- Bladder neck stricture is under 2% after RARP versus up to 16% after open surgery, because of precise mucosa-to-mucosa anastomosis.

2. RAPN (robot-assisted partial nephrectomy): compared with laparoscopy it gives **fewer conversions, shorter warm ischaemia**, smaller eGFR (estimated glomerular filtration rate) loss and shorter stay. **Trifecta** = negative margin, no major complication and warm ischaemia ≤ 25 minutes. Firefly ICG (indocyanine green) aids selective clamping.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018): non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), less blood loss and fewer transfusions.
- **iROC:** RARC with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, and fewer thromboembolic and wound complications.

4. Reconstruction: pyeloplasty, ureteric reimplantation, and robotic BMG (buccal mucosa graft) ureteroplasty.

5. Others: adrenalectomy, R-RPLND (robotic retroperitoneal lymph node dissection), simple prostatectomy, sacrocolpopexy, paediatric reconstruction and **RAKT (robot-assisted kidney transplantation)**.

Training and credentialing

- Pathway: didactics, **VR (virtual reality) simulators**, dry and wet lab, then proctored console time.
- **FRS (Fundamentals of Robotic Surgery):** a multicentre RCT showed better speed and accuracy after training. Simulator scores correlate with live RARP performance.
- Basic proficiency comes after about 40 robotic cases, but outcomes keep improving with volume.

Complications specific to robotics

Problem	Details
Steep Trendelenburg (about 22-25° or more)	Facial and laryngeal oedema (delay extubation), raised IOP (intraocular pressure), corneal abrasion, rarely ischaemic optic neuropathy
Positional nerve injury (up to ~6% in robotic series)	Brachial plexus (shoulder braces, arm abduction >90°): avoid braces; femoral nerve (hip flexion)
Compartment syndrome	Long lithotomy; fasciotomy if pressure >30 mmHg with symptoms
Electrosurgical bowel injury	Insulation failure and capacitive coupling; may present late
Robot malfunction	About 0.4%; rarely forces conversion
Port-site hernia	Fewer with transverse incisions

Recent advances

- **Single-port** extraperitoneal and transvesical surgery, with same-day discharge.
- **AI (artificial intelligence)**: video-based skill assessment and outcome prediction.
- **Telesurgery**: long-distance operations reported since 2023-25; latency is the main limit.

Diagram to draw: da Vinci block diagram (console, vision cart, patient cart) and RARP port placement.

Key points to remember:

- da Vinci approved 2000; Si 2009 (dual console), Xi 2014, SP 2018 (one 2.5-cm port); patents expired 2019.
- Yaxley 2016 and Coughlin 2018: equal function at 12 weeks and 24 months; surgeon matters more than platform.
- RAZOR: non-inferior 2-year PFS; iROC: intracorporeal RARC gives more days alive and out of hospital.
- Steep Trendelenburg causes facial and laryngeal oedema and raised IOP; avoid shoulder braces.

IV-2. Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation

Type: LAQ (25 marks)

Introduction

- The organ shortage has led to wider use of **marginal (extended criteria) donors**. Their kidneys carry a higher risk of DGF (delayed graft function), primary non-function and shorter graft survival, but transplantation still beats staying on dialysis.

Part A: Marginal kidney donor

Definitions and categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002

- Donor aged **≥60 years**, or **50-59 years** with **at least 2 of: hypertension, terminal creatinine >1.5 mg/dL, death from CVA (cerebrovascular accident)**.
- About 70% higher relative risk of graft loss than SCD (standard criteria donor) kidneys.

2. KDPI (Kidney Donor Profile Index)

- Percentile of the KDRI (Kidney Donor Risk Index), built from 10 donor factors. **KDPI >85%** replaced the ECD label in the US (December 2014).
- US allocation pairs the best kidneys (low KDPI) with recipients of the longest expected survival (EPTS (Estimated Post-Transplant Survival)): **longevity matching**.

3. DCD (donation after circulatory death): Maastricht classification

Category	Description	Type
I	Dead on arrival	Uncontrolled
II	Unsuccessful resuscitation	Uncontrolled
III	Awaiting cardiac arrest after withdrawal of therapy	Controlled (commonest)
IV	Cardiac arrest in a brain-dead donor	Controlled or uncontrolled
V	Euthanasia	Controlled

- Warm ischaemia demands rapid aortic cannulation and cooling. DCD kidneys have **higher DGF (roughly 40-50%)** but **similar long-term graft survival**.

4. Other marginal donors

- AKI (acute kidney injury)** donors: do not discard on terminal creatinine alone.
- HCV (hepatitis C virus)-viraemic donors to HCV-negative recipients**: DAAs (direct-acting antivirals) give near-universal cure.
- Small paediatric donors (<15 kg)**: en bloc transplant, with survival comparable to adult kidneys.
- Marginal living donors** (older, controlled hypertension): KDIGO (Kidney Disease: Improving Global Outcomes) 2017 advises a risk-based approach.

Assessment and optimisation

- Remuzzi biopsy score**: glomeruli, tubules, interstitium and arteries each scored 0-3 (total 0-12).

Score	Action
0-3	Single transplant
4-6	Dual kidney transplant (two kidneys give enough nephron mass)

Score	Action
≥7	Discard

- **HMP (hypothermic machine perfusion)** extends preservation and **reduces DGF**. **Moers et al., NEJM 2009** showed less DGF (about 21% vs 27%), most benefit in ECD kidneys. Campbell notes no difference in long-term graft survival.
- **NMP (normothermic machine perfusion)** allows viability assessment; clinical experience is limited.
- **Recipient matching**: the **ESP (Eurotransplant Senior Program)** gives kidneys from donors **≥65 years** to recipients **≥65 years**, prioritising short cold ischaemia over HLA (human leucocyte antigen) matching. An elderly recipient may gain more from a quicker older kidney than from waiting. **Informed consent** is essential.

Outcomes and care

- **Ojo et al. (JASN 2001)**: ECD transplantation improves survival over dialysis, though graft survival is lower than with standard kidneys.
- With DGF, use antibody induction and delay nephrotoxic CNI (calcineurin inhibitor) therapy.

Part B: Newer immunosuppressant drugs

Current standard

- Over 90% receive induction, and over 90% receive **tacrolimus + mycophenolate** (± steroids). Acute rejection has fallen below 10%.
- The unsolved problem is chronic **AMR (antibody-mediated rejection)**. Newer agents aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

Induction agents

Drug	Mechanism	Key points
Basiliximab	Anti-CD25 (IL-2 (interleukin-2) receptor α), non-depleting	20 mg IV (intravenous) on day 0 and day 4 ; low-risk patients
rATG (rabbit antithymocyte globulin)	Polyclonal T-cell depletion	High risk or DGF; risks CMV (cytomegalovirus) and PTLD (post-transplant lymphoproliferative disorder)

Maintenance agents

Drug	Mechanism	Key points
Extended-release tacrolimus	Calcineurin inhibition	Once daily, better adherence
mTOR (mammalian target of rapamycin) inhibitors : sirolimus, everolimus	Arrest the T-cell cycle	Not nephrotoxic; fewer CMV/BK infections. TRANSFORM (JASN 2018) : everolimus with reduced CNI was non-inferior . Cause poor wound healing and lymphocele
Belatacept	CTLA4-Ig (cytotoxic T-lymphocyte antigen 4-immunoglobulin) ; blocks CD28 costimulation (signal 2)	FDA (Food and Drug Administration) 2011; 5 mg/kg every 4 weeks maintenance. BENEFIT: better GFR (glomerular filtration rate) than ciclosporin but more early rejection . Contraindicated in EBV (Epstein-Barr virus)-seronegative recipients (PTLD)

Desensitisation and AMR

Drug	Mechanism	Use
IVIG (intravenous immunoglobulin), plasmapheresis, rituximab	Remove or modulate antibody	First-line AMR combinations
Eculizumab	Anti-C5 complement	Selected AMR; prevents aHUS (atypical haemolytic uraemic syndrome) recurrence; meningococcal vaccination mandatory
Anti-IL-6 therapy	Blocks IL-6	Newer option for chronic active AMR
Imlifidase	Enzyme that cleaves IgG within hours	EMA (European Medicines Agency) conditional approval 2020 for highly sensitised, crossmatch-positive patients

- **Voclosporin** is approved for lupus nephritis, **not for transplantation**.

Side-effect summary

Drug	Signature toxicity
Tacrolimus	Nephrotoxicity, NODAT (new-onset diabetes after transplant), tremor
Mycophenolate	GI (gastrointestinal) upset, teratogenicity
Belatacept	PTLD, early rejection

Recent advances

- **dd-cfDNA (donor-derived cell-free DNA)**: negative predictive value over 90% for rejection, avoiding unnecessary biopsy.
- Normothermic perfusion with ex vivo repair.
- **Pig-kidney xenotransplantation** into a living human (MGH (Massachusetts General Hospital), March 2024).

Diagram to draw: three-signal T-cell activation model, marking belatacept (signal 2) and basiliximab/mTOR inhibitors (signal 3).

Key points to remember:

- ECD (UNOS 2002): age ≥ 60 , or 50-59 with ≥ 2 of hypertension, creatinine >1.5 mg/dL, CVA death; replaced by KDPI $>85\%$ (December 2014).
- Remuzzi 0-3 single, 4-6 dual, ≥ 7 discard; HMP reduces DGF (Moers 2009); ESP matches ≥ 65 donors to ≥ 65 recipients.
- Belatacept: better GFR, more early rejection, contraindicated if EBV-seronegative; TRANSFORM: everolimus with low CNi non-inferior.
- Imlifidase (EMA 2020) for highly sensitised patients; anti-IL-6 for chronic AMR; dd-cfDNA for monitoring.

IV-3a. Etiology and management of hypospadias fistula

Type: SAQ (10 marks)

Definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak anywhere below the meatus. It is the **commonest complication of hypospadias repair**: just under 10% at short-term follow-up.
- **TIP (tubularised incised plate) meta-analysis (Pfistermuller 2015)**: fistula 5.7%. After proximal repairs it is 3–45%.

Aetiology (risk factors)

- **Distal obstruction** (meatal stenosis, stricture) causes high pressure and turbulence behind the repair. It is the **most important correctable cause**.
- Ischaemia, oedema, infection, haematoma and tension.
- Technical: **overlapping suture lines**, poorly inverted epithelium, slowly absorbable sutures, **no barrier layer**.
- Patient: proximal meatus, glans width <14 mm, previous surgery.

Prevention

- A **vascularised barrier layer** (dartos, or **tunica vaginalis** in proximal or redo cases), offset suture lines, tension-free repair.

Evaluation

- Presents as a split or double stream, mostly within 2 years.
- Under anaesthesia: **calibrate** with a bougie à boule, **urethroscopy** if calibre is irregular, and a **"stress test"** (inject saline or methylene blue with proximal and distal compression) to reveal hidden tracts.

Management

- Small early fistulae rarely close spontaneously, so **wait at least 6–12 months** for tissues to soften.
- **Always correct distal obstruction at the same sitting.**

Fistula	Repair
Small, shaft	Excise tract, subepithelial closure, multiple offset layers with dartos barrier
Large, or thin tissue	Trap-door or island flap of shaft skin plus vascularised second layer
Coronal margin (glans dehiscence)	Revise urethroplasty and glans, not simple excision

- Recurrence is about **20%** even after good repair; interposed vascularised layers reduce it.
- Catheter: none for simple repairs; 7–14 days for complex redo repairs.

UCF → wait 6–12 months → calibrate, urethroscopy, stress test
└ distal obstruction → correct at same sitting
└ small shaft → excise + layered closure + barrier flap
└ large/coronal → flap repair or redo urethroplasty

Recent advances

- Standardised complication definitions; objective scores such as **HOPE (Hypospadias Objective Penile Evaluation)**.

Diagram to draw: cross-section of neourethra, dartos barrier and skin with offset suture lines.

Key points to remember:

- Commonest complication: just under 10%; 5.7% after TIP (Pfistermuller 2015); up to 45% after proximal repair.
 - Always exclude and correct distal obstruction.
 - Wait 6–12 months; layered closure, barrier flap, offset suture lines; recurrence about 20%.
 - A coronal fistula means glans dehiscence and needs revision of urethroplasty and glans.
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IV-3b. Silodosin

Type: SAQ (10 marks)

Definition

- Silodosin is the **most α 1A-selective adrenoceptor antagonist**, used mainly for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).

Pharmacology

- Selectivity for α 1A is well over 100-fold that for α 1B (162 : 1 commonly quoted) and about 50-fold that for α 1D.
- Sparing vascular α 1B receptors means **minimal hypotension**. Strong α 1A blockade of the vas and seminal vesicles causes **failure of emission (anejaculation)**.
- Onset: flow improves within 2–6 hours, symptoms within 3–4 days. Metabolised by **CYP3A4**.

Dose and contraindications

Item	Detail
Dose	8 mg once daily (OD) with a meal
CrCl (creatinine clearance) 30–50 mL/min	4 mg daily
Contraindicated	CrCl <30 , severe liver impairment, strong CYP3A4 inhibitors

Efficacy

- European trial (Chapple, Eur Urol 2011): non-inferior to tamsulosin 0.4 mg.** Silodosin is **not more effective** than tamsulosin 0.4 mg.
- Gives the greatest fall in BOOI (bladder outlet obstruction index) of the α 1-blockers.
- It does not shrink the prostate; add a 5-ARI (5 α -reductase inhibitor) for large glands.

Adverse effects

- Anejaculation** is the signature effect: over a quarter of men in trials (about 14% vs 2% with tamsulosin head-to-head). Few stop the drug, and men with anejaculation had better responses.
- No excess hypotension, even in the elderly or on antihypertensives.
- IFIS (intraoperative floppy iris syndrome)**: not specifically studied, but reported; warn before cataract surgery.

Uses

- BPH/LUTS, especially elderly and cardiac patients.
- MET (medical expulsive therapy)** for distal ureteric stones 5–10 mm: meta-analyses show **faster and higher expulsion than tamsulosin**. The SUSPEND trial (2015) showed no overall benefit for tamsulosin.
- Ureteric stent symptoms; CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome) of recent onset.

Comparison of α -blockers

Drug	Hypotension	Ejaculatory dysfunction
Silodosin	Lowest	Highest

Drug	Hypotension	Ejaculatory dysfunction
Tamsulosin	Low	Moderate
Alfuzosin	Low	Lowest of uroselective drugs
Doxazosin, terazosin	Highest	Low

Diagram to draw: $\alpha 1$ subtypes: $\alpha 1A$ prostate and vas, $\alpha 1B$ vessels, $\alpha 1D$ bladder.

Key points to remember:

- Most $\alpha 1A$ -selective; 8 mg OD with a meal, 4 mg if CrCl 30–50, avoid if CrCl <30.
- Equal to tamsulosin 0.4 mg, not better; greatest BOOI reduction.
- Anejaculation (failure of emission) is the signature effect; hypotension is the lowest.
- Better than tamsulosin for MET of distal stones in meta-analyses.

IV-3c. Sacral neuromodulation

Type: SAQ (10 marks)

See Paper III, Q3a (identical question). Recent advances to add:

- **Rechargeable bladder neurostimulators** (Axonics and Medtronic, FDA (Food and Drug Administration) 2019), and long-life non-rechargeable devices lasting ≥ 10 years (2021).
 - **Whole-body MRI (magnetic resonance imaging)-conditional** systems at 1.5 T and 3 T.
 - **Implantable tibial nerve stimulators** (BlueWind, eCoin) avoid repeated clinic visits for tibial stimulation.
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IV-3d. Urethral stents

Type: SAQ (10 marks)

Definition

- A tubular endoprosthesis placed endoscopically in the **prostatic or bulbar urethra** to keep it open, mainly in men unfit for definitive surgery.

Classification

Type	Examples	Features
Temporary, non-epithelialising	Memokath (thermo-expandable nitinol), Allium	Designed to avoid epithelial incorporation, so removable; left 6 months to 1 year
Permanent, epithelialising	UroLume	Covered by urothelium, so hard to remove; withdrawn
Biodegradable	Experimental	Dissolve

Indications

- BPH (benign prostatic hyperplasia) with retention in men unfit for TURP (transurethral resection of prostate).
- Recurrent short bulbar stricture** in men unfit for urethroplasty.
- DSD (detrusor-sphincter dyssynergia)** in spinal cord injury, as a reversible alternative to sphincterotomy. Campbell now regards stents for DSD as **abandoned** because of complications.

Technique

- Local or regional anaesthesia with endoscopic or fluoroscopic guidance. Memokath expands when warmed and softens when cooled for removal.
- A bulbar stent must **not cross the external sphincter**, or incontinence results.

Complications

Complication	Comment
Migration	Commonest with temporary stents
Encrustation and stones	Especially prostatic stents
Hyperplastic tissue ingrowth	UroLume; causes restenosis
Incontinence	Stent crossing the sphincter
Pain, painful erections	Bulbar stents

Results

- Prostatic stents free most unfit men from catheters in the short term, but many are later removed.
- Removable bulbar stents succeed only in small series; Memokath results are variable.
- No stent removes spongiositis, so stents cannot cure a stricture.

Guideline position

- AUA (American Urological Association): no stent in a man considering urethroplasty**, because it makes reconstruction harder. Urethroplasty remains the standard.

Recent advances

- **Optilume** paclitaxel-coated balloon (inhibits fibroblasts): **ROBUST III**, after failed endoscopic treatments, stricture-free about 75% vs 27% with endoscopic management.
- **iTind** (temporary implantable nitinol device) for BPH: left 5–7 days; AUA 2023 option for 25–75 mL glands without an obstructing median lobe.

Diagram to draw: sagittal urethra with a prostatic stent and a bulbar stent distal to the sphincter.

Key points to remember:

- Temporary (Memokath, Allium) vs permanent epithelialising (UroLume, withdrawn).
 - Indications: unfit BPH, unfit bulbar stricture, DSD (now largely abandoned).
 - Main problems: migration, encrustation and hyperplastic ingrowth.
 - AUA: no stent if urethroplasty is considered; Optilume (ROBUST III) and iTind are the recent advances.
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IV-3e. Studer Neobladder

Type: SAQ (10 marks)

Introduction

- An **orthotopic, low-pressure ileal reservoir** with a long afferent limb, joined to the urethra (**Studer, Bern, 1989**).

Principles

- **Detubularisation** abolishes coordinated contractions, keeping pressure low.
- A **spherical shape** (Laplace's law) gives the greatest volume for a given surface area.
- The **isoperistaltic afferent limb** limits reflux and accepts short ureters.
- Patients void by **abdominal straining with pelvic floor relaxation**. Continence depends on the intact **rhabdosphincter** (pudendal nerve).

Technique

1. Isolate **54-56 cm of ileum**, ending **15-25 cm proximal to the ICV (ileocaecal valve)** to preserve B12 and bile-salt absorption.
2. Open the **distal 40-44 cm** (two 20-22-cm limbs) on the antimesenteric border and fold into a U; keep the **proximal 10-15 cm** tubular as the **afferent limb**.
3. Fold again into a sphere; **direct (refluxing) spatulated ureteroileal anastomoses**, stented.
4. Join the **most dependent point** to the urethra. - Capacity grows from under 200 mL to **500-600 mL** within a year.

Selection

- **Absolute contraindications**: cancer at the urethral margin; significant urethral stricture; severe renal or hepatic dysfunction; inability to self-catheterise.
- **Renal threshold**: eGFR (estimated glomerular filtration rate) **>35-40 mL/min** (creatinine <1.7-2.2 mg/dL).

Postoperative care

- Mucus irrigation, oral bicarbonate, cystogram before catheter removal.
- Timed voiding; **night alarm 2-3 times**.

Outcomes

- **Daytime continence 80-90%** (mostly by 3-6 months); **nocturnal incontinence 20-50%**.
- Self-catheterisation needed in **10-20% of men and about 30% of women**; men leak, women retain.
- It refluxes on cystogram, yet kidneys are preserved in about 95%.
- **USC (University of Southern California) RCT**: Studer equalled the antireflux T pouch for renal function and infection, with fewer reoperations.

Complications

- **Early**: urine leak, ileus, infection, mucus retention.
- **Late**: **hyperchloraemic metabolic acidosis, B12 deficiency, ureteroileal stricture (<3%)**, stones, retention, pyelonephritis.

Recent advances

- Intracorporeal robotic Studer neobladder; uterus- and nerve-sparing cystectomy in women.

Diagram to draw: marked ileal segment, then the spherical pouch with afferent limb.

Key points to remember:

- About 54–56 cm ileum: 40–44 cm reservoir (U then sphere) + 10–15 cm afferent limb; spare the terminal 15–25 cm.
 - Detubularisation plus a sphere gives low pressure and high capacity (500–600 mL by 1 year).
 - Day continence 80–90%; night incontinence 20–50%; CISC 10–20% men, ~30% women.
 - Contraindicated if eGFR <35–40 mL/min, urethral margin positive or unable to self-catheterise.
-

IV-3f. Cryotherapy for carcinoma prostate

Type: SAQ (10 marks)

Definition

- **Image-guided transperineal** ablation of prostate cancer by extracting heat to lethal cold. The transperineal route suits **anterior** tumours (HIFU (high-intensity focused ultrasound) is transrectal).

Physical principle

- **Joule-Thomson effect:** high-pressure **argon** expands at the probe tip to freeze; **helium** actively thaws.
- **Thermocouples** monitor the sphincter and neurovascular bundles; a **urethral warmer** prevents urethral sloughing.

Generations

Generation	Features
1st	Liquid nitrogen, no imaging; many fistulae
2nd	TRUS (transrectal ultrasound), urethral warmer
3rd	Argon/helium, 17-gauge probes
4th	Focal therapy with MRI (magnetic resonance imaging) fusion

Mechanisms of cell death

1. **Intra- and extracellular ice** and recrystallisation disrupt membranes.
2. **Vascular stasis and ischaemia**, the main effect at the margins.
3. Apoptosis and immune-mediated effects. - Aim for **-40 °C**, fast freeze, slow thaw and **two freeze-thaw cycles**. The ice-ball edge is sublethal, so ice must extend beyond the tumour.

Indications

1. **Salvage after radiation failure** (main role): biopsy-proven local recurrence, no metastases, **PSA (prostate-specific antigen) <10 ng/mL (ideally <4)**, doubling time ≥ 16 months.
2. **Primary whole-gland:** men unfit for surgery or radiotherapy.
3. **Focal or hemi-gland:** MRI-visible unilateral grade group 2-3 lesion.

Outcomes

- Salvage: biochemical control about **55-71% at 3-5 years** (Phoenix: nadir + 2 ng/mL).
- **Donnelly et al., Cancer 2010** (RCT): **non-inferior** to external beam radiotherapy; worse sexual function.
- Focal: the COLD (Cryo On-Line Data) registry shows better erectile recovery than whole-gland.

Complications

Complication	Primary / focal	Salvage
ED (erectile dysfunction)	Almost always (whole gland); often kept (focal)	Very common
Incontinence	Rare (focal pad-free 95-100%)	Now single digit
Rectourethral fistula	$\leq 0.8\%$ (focal)	About 1-3.4%

Guideline positions

- **EAU (European Association of Urology):** primary or focal cryotherapy **only within trials or registries**; salvage is an option.

Recent advances

- MRI-TRUS fusion focal cryotherapy; PSMA (prostate-specific membrane antigen) PET to select salvage patients.

Diagram to draw: sagittal prostate with ice ball, -40°C isotherm inside the 0°C edge.

Key points to remember:

- Argon freezes and helium thaws (Joule-Thomson); -40°C , two freeze-thaw cycles, urethral warmer.
 - Main role is salvage after radiation: PSA <10 (ideally <4), doubling time ≥ 16 months.
 - Whole-gland ED is almost universal; fistula $\leq 0.8\%$ focal, 1–3.4% salvage.
 - EAU: primary or focal cryotherapy only within trials or registries; Phoenix = nadir + 2.
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Rapid Revision Sheet - Paper IV

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Yaxley 2016 and Coughlin 2018: equal function at 12 weeks and 24 months; surgeon matters more than platform.
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- Imlifidase (EMA 2020) for highly sensitised patients; anti-IL-6 for chronic AMR; dd-cfDNA for monitoring.

IV-3a. Etiology and management of hypospadias fistula (SAQ)

- Commonest complication: just under 10%; 5.7% after TIP (Pfistermuller 2015); up to 45% after proximal repair.
- Always exclude and correct distal obstruction.
- Wait 6–12 months; layered closure, barrier flap, offset suture lines; recurrence about 20%.
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IV-3b. Silodosin (SAQ)

- Most $\alpha 1A$ -selective; 8 mg OD with a meal, 4 mg if CrCl 30–50, avoid if CrCl <30 .
- Equal to tamsulosin 0.4 mg, not better; greatest BOOI reduction.
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IV-3d. Urethral stents (SAQ)

- Temporary (Memokath, Allium) vs permanent epithelialising (UroLume, withdrawn).
- Indications: unfit BPH, unfit bulbar stricture, DSD (now largely abandoned).
- Main problems: migration, encrustation and hyperplastic ingrowth.
- AUA: no stent if urethroplasty is considered; Optilume (ROBUST III) and iTind are the recent advances.

IV-3e. Studer Neobladder (SAQ)

- About 54–56 cm ileum: 40–44 cm reservoir (U then sphere) + 10–15 cm afferent limb; spare the terminal 15–25 cm.
- Detubularisation plus a sphere gives low pressure and high capacity (500–600 mL by 1 year).
- Day continence 80–90%; night incontinence 20–50%; CISC 10–20% men, ~30% women.

- Contraindicated if eGFR <35–40 mL/min, urethral margin positive or unable to self-catheterise.

IV-3f. Cryotherapy for carcinoma prostate (SAQ)

- Argon freezes and helium thaws (Joule–Thomson); -40°C , two freeze–thaw cycles, urethral warmer.
- Main role is salvage after radiation: PSA <10 (ideally <4), doubling time ≥ 16 months.
- Whole-gland ED is almost universal; fistula $\leq 0.8\%$ focal, 1–3.4% salvage.
- EAU: primary or focal cryotherapy only within trials or registries; Phoenix = nadir + 2.